A thematic portfolio for Swissquote (work in progress)

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Abstract

While Swissquote already provides a broad selection of thematic portfolios targeting growing industries with seemingly considerable (albeit risky) growth potential, there is no theme covering an industry with near-growth certainty. Our portfolio, *Vital Infrastructur*e, caters to the portion of investors currently neglected in Swissquote's stock bundles: we provide a bundle with low risk, high stability, and potential for growth. We have curated companies that will appeal to any investor interested in placing their money in a low-risk environment, with the promise of steady growth resulting from increasing demands from a growing population.

1 Introduction

Thematic portfolios comprise a carefully selected assortment of stocks of a particular domain or industry, handpicked by domain experts. Clients opting to invest in these thematic portfolios are looking ahead and can therefore position their assets for faster growth and higher returns. The advantages of a thematic portfolio are as follows:

1. Simplicity & transparency: A broad selection of assets can be traded simultaneously, and it is transparent which domains or industries funds are being invested in.
2. Intuitiveness: Current trends can be traded simply by customers without the need for time, knowledge or funds to build a personal portfolio.
3. Insightful Ideas: Thematic portfolios provide access to innovating investment ideas, while allowing a broad diversification.
4. Value alignment: A common understanding of today's world is the client's greatest advantage. Thematic portfolios enable clients to invest in ready-made instruments that align with their beliefs.

2 Related work

3 Methods

3.1 Theme selection

The companies we have selected fall under the theme of vital infrastructure, meaning companies that provide essential social services necessary for any cohesive population to survive in its day-to-day life. We are specifically interested in energy systems (generation, transmission, storage), logistics networks (multi-modal transport, warehousing), and resource management (water utilities, waste processing), as we expect these areas to experience the most significant growth in the long term by benefiting from the mega-trends of digitalisation and globalisation while remaining insulated from unpredictable demand shocks. This theme is expected to maintain greater stability due to:

1. Systematic demand: Any population will inevitably require firms to provide services for energy, logistics, and resource management. As such, we have selected companies from our basket that hold a strong position in these fields, either by being a generally large and dominant market player or by being smaller and more innovative within a specific niche.
2. Demand inelasticity: Even in the face of sudden economic disruptions, demand for energy and water/waste management is expected to remain essentially unchanged. This can be easily verified by observing the stock prices of companies operating in these industries during the COVID era, where comparatively fewer losses were incurred.

Furthermore, we are confident in the growth potential of Vital Infrastructure despite its seemingly dull nature. We believe so as:

3. Benefits of digitalisation: While many companies are forcefully integrating new technologies, such as AI, into their business models, our selected companies will naturally enjoy significant efficiency gains. Choosing how to allocate resources, reacting quickly to demand shifts, and collecting information for routing optimisations. These activities are all set to experience significant performance improvements in the coming decade. Waste and water management, energy grids, and logistics companies, therefore, have significant growth potential.
4. Benefits of global supply chains: Despite the seemingly growing political instability in the West, our team stands firmly with the position that the world is becoming increasingly interconnected, with the economic value from international trade being lucrative enough to be considered an inevitability, whether ethically good or not. As such, our infrastructure to facilitate economic trade must continue to be maintained, innovated and expanded.

To invest in infrastructure is to believe in holistic, long-term global productivity, which is shown to be growing much faster than at any other point in history. With our stock bundle, Swissquote can offer its users a new theme that embodies a lower-risk profile alongside sustainable and predictable returns for investors.

3.2 Stock selection

After deciding on the specific theme we chose stocks for each of the subcategories: energy systems, logistics networks, and resource management. The number of stocks chosen per category is an open question. We list the individual reasoning for selction for each stock in section 7.1.

3.3 Stock weighting

The weighting of the stocks in the portfolio was done using volatility. This method aims at minimising potential drawdown and risk. Concretely, the weight of each stock is based on the inverse of its volatility in the past financial year (≈ 250 days). This enables equal risk contribution. Lowering the weight of highly volatile companies enables the thematic portfolio to outperform portfolios with fixed weightings by exhibiting less volatility and lower drawdowns.

This method might seem basic at first. One might wish for a more complex method to "perfectly" predict future risk associated with each stock. To the best of our knowledge, such a method does not exist. The types of risk that can influence a market include liquidity risk, market risk, inflation risk, interest rate risk, geopolitical risk, and others. These risks interact with each other over time and fluctuate significantly, making it challenging to develop a method to precisely gauge the future impact of individual stocks within the thematic portfolio.

However, price volatility can be used to measure the risk associated with stocks reliably. Historical price volatilities are more correlated over time than historical price returns. This effect is known as heteroscedasticity (Omran & McKenzie, 2000).

To make the calculations, we used open-source data from Yahoo Finance. Let n be the number of stocks and T the number of days of available stock price data. We define $X_{i,j}$ to be the price of stock $i = 1, \ldots, n$ and on day $j = 1, \ldots, T$. The following steps were done:

1. Import data using the yfinance API
2. Clean data for stock splits and missing data points. When working with recent data from Yahoo Finance, the data is often incomplete.
3. For each stock $i = 1, \ldots, n$ individually:
   (a) Calculate the fractional change between consecutive days for the whole timeframe:
   $$r_{i,j} = \frac{X_{i,j} - X_{i,j-1}}{X_{i,j-1}}, \quad j = 2, \ldots, T$$
   (b) Calculate the sum over all fractional changes:
   $$\bar{r}_i = \frac{1}{T-1} \sum_{j=2}^{T} r_{i,j}$$
   (c) Calculate the standard deviation of the fractional changes (a metric of volatility):
   $$\sigma_i = \sqrt{\frac{1}{T-1} \sum_{j=2}^{T} (r_{i,j} - \bar{r}_i)^2}$$
   (d) Calculate the inverse volatility:
   $$v_i = \frac{1}{\sigma_i}$$
4. Calculate the sum over all inverse volatilities:
   $$V = \sum_{i=1}^{n} v_i$$
5. For each stock individually:
   Divide the inverse volatility by the sum of all inverse volatilities. This will give a unique weighting of each stock in the portfolio:
   $$w_i = \frac{\frac{1}{\sigma_i}}{\sum_{i=1}^{n} v_i}$$

We can summarise steps 3-5 as follows in one expression:

$$w_i = \frac{\left(\sqrt{\frac{1}{T-2} \sum_{j=2}^{T} \left(\frac{X_{i,j}-X_{i,j-1}}{X_{i,j-1}} - \frac{1}{T-1} \sum_{k=2}^{T} \frac{X_{i,k}-X_{i,k-1}}{X_{i,k-1}}\right)^2}\right)^{-1}}{\sum_{\ell=1}^{n} \left(\sqrt{\frac{1}{T-2} \sum_{j=2}^{T} \left(\frac{X_{\ell,j}-X_{\ell,j-1}}{X_{\ell,j-1}} - \frac{1}{T-1} \sum_{k=2}^{T} \frac{X_{\ell,k}-X_{\ell,k-1}}{X_{\ell,k-1}}\right)^2}\right)^{-1}}$$

Summing, all calculated weightings yield a value of one, indicating that the weighting is a valid probability distribution. This entire process could also be completed using data from Swissquote, which would hopefully not contain missing data points.

For the portfolio breakdown in section 7.1, we provide the weighting based on the 2024 financial year.

To summarise, the selection of a stock portfolio was done qualitatively, and the actual weighting of the stocks in the chosen portfolio was determined quantitatively. This approach strikes a good balance between solely subjective weighting of the stocks, which is prone to human error, and solely examining historical stock data as time series data. The code can be found on our GitHub repository, along with a step-by-step explanation.

4 Results

5 Discussion

In real-world implementation, we would adjust the portfolio's weight on a daily, weekly, or monthly basis, taking balancing costs into account. Additionally, one could adapt the lookback period from 250 days to, for example, 6 months or 3 months. If the weighting were to be periodically reevaluated, a threshold could be set so that only changes that justify the transaction costs are implemented.

6 Conclusion

Our investment theme, *Vital Infrastructure*, is based on the idea that specific systems, such as energy, water, waste management, and logistics, form essential components of modern society. These sectors address fundamental needs and are expected to remain relevant and necessary over the long term, regardless of economic cycles or shifting trends.

Rather than focusing on disruptive technologies or high-risk innovation, our strategy targets companies that support and improve the operation of these critical systems. These firms often provide the tools and technologies, such as software, automation, and analytics, that enable incremental yet meaningful improvements in efficiency, reliability, and sustainability. To build the portfolio, we selected companies that fit the theme in both substance and role. We then applied a volatility-aware weighting strategy to help manage risk while maintaining exposure to promising opportunities.

For investors, this theme offers a thoughtful balance: exposure to stable, long-term demand with the potential for growth driven by ongoing modernisation. It is a pragmatic investment approach grounded in societal need, one that prioritises resilience and steady value creation over speculation.

7 Appendix

7.1 Portfolio breakdown

In this chapter we show the detailed breakdown of the portfolio into its components. In the first table we provide the name ISIN and description of the company. In the second table we provide the reason for inclusion for each individual company. In the third table, we provide the weighting of each company using the method introduced in section 3.3.

Company	ISIN	Description
Ecolab Inc (ECL)	US2788651006	Global leader in water treatment and hygiene solutions, serving industries such as food, healthcare, and manufacturing.
Badger Meter Inc (BMI)	US0565251081	Specialises in smart water meters and flow measurement technologies for utilities, municipalities, and industries.
Veralto Corporation (VLTO)	US92338C1036	Provides water quality solutions, including chemical treatment services and UV-filtration systems.
Xylem Inc (XYL)	US98419M1009	Offers pumps, smart meters, filtration, and wastewater systems across residential, industrial, and utility sectors.
Georg Fischer AG (GF)	CH1169151003	Swiss leader in advanced piping systems for safe transport of water, chemicals, and gases.
Waste Management Inc (WM)	US94106L1098	North American leader in waste collection, recycling, landfill operations, and renewable energy generation.
GFL Environmental Inc (GFL)	CA36168Q1046	Canadian company offering waste, recycling, liquid waste management, and soil remediation services.
Clean Harbors Inc (CLH)	US1844961078	Specialises in hazardous waste disposal, emergency spill response, and used oil recycling.
Veolia Environnement (VIE.PA)	FR0000124141	French multinational in water, waste, and energy services, operating in 50+ countries.
Casella Waste Systems (CWST)	US1474481041	Regional U.S. company in waste, recycling, organics, and resource recovery.
Cleanaway Waste Mgmt (CWY.AX)	AU000000CWY3	Australia's largest waste management company with nationwide infrastructure.
Tomra Systems ASA (TOM.OL)	NO0005668905	Develops sorting and reverse vending technologies for recycling and waste.
Symbotic Inc (SYM)	US87151X1019	Builds AI-powered robotic warehouse automation systems.
GXO Logistics Inc (GXO)	US36262G1013	Major provider of outsourced warehousing, distribution, and e-commerce fulfilment.
Autostore Holdings Ltd (AUTO.OL)	BMG0670A1099	Robotics company specialising in Cube Storage Automation systems.
Zebra Technologies (ZBRA)	US9892071054	Provides barcode scanners, RFID, rugged computers, and enterprise asset intelligence tools.
KION GROUP AG (KGX.DE)	DE000KGX8881	Offers forklifts, warehouse equipment, and supply chain automation via Dematic.
Cognex Corporation (CGNX)	US1924221039	Provides machine vision systems for inspection, identification, and robot guidance.
Manhattan Associates (MANH)	US5627501092	Develops supply chain and commerce software, incl. WMS, TMS, and OMS.
BKW AG (BKW)	CH0130293662	Swiss energy and infrastructure firm focused on sustainable power and grids.
Enel (ENEL.MI)	IT0003132476	Italian energy giant with global operations in renewables and grid management.
NextEra Energy (NEE)	US65339F1012	U.S. leader in renewable power generation and modern grid infrastructure.
Iberdrola (IBE.MC)	ES0144580Y14	Spanish multinational focused on renewables, grids, and distribution.
Ørsted (ORSTED.CO)	DK0060094928	Global leader in offshore wind and smart grid technology.
First Solar (FSLR)	US3364331070	Provides advanced solar modules and PV energy systems.
Siemens Energy (ENR.DE)	DE000ENER6Y0	Develops advanced power transmission and renewable grid technology.

Company	Reason for inclusion
Ecolab Inc	Focus on water treatment and sustainability, with a diverse and resilient client base.
Badger Meter Inc	Provides smart metering solutions that improve water efficiency and leak detection.
Veralto Corporation	Strong positioning in water quality and monitoring, benefitting from stricter regulations.
Xylem Inc	Broad portfolio covering transportation and treatment, supported by strategic acquisitions.
Georg Fischer AG	Provides reliable piping systems addressing critical infrastructure needs.
Waste Management Inc	Market leader with pricing power, long-term contracts, and renewable energy initiatives.
GFL Environmental Inc	High-growth ESG player expanding aggressively in waste solutions.
Clean Harbors Inc	Specialist in hazardous waste and regulatory-driven environmental services.
Veolia Environnement	Diversified global leader in environmental services, strengthened by Suez merger.
Casella Waste Systems	Strong regional presence with innovative waste-to-energy practices.
Cleanaway Waste Mgmt	Dominant Australian player investing in circular waste systems.
Tomra Systems ASA	Tech leader in sensor-based recycling and reverse vending solutions.
Symbotic Inc	Automates warehouses with AI robotics, enhancing supply chain resilience.
GXO Logistics Inc	Pure-play logistics operator driving efficiency with automation.
Autostore Holdings Ltd	Provides high-density automated storage solutions for e-commerce growth.
Zebra Technologies	Supplies essential data capture and tracking technology for automation.
KION GROUP AG	Dual focus on forklifts and warehouse automation positions it strongly in logistics.
Cognex Corporation	Provides machine vision, a key enabling tech for automation and quality control.
Manhattan Associates	Software "brain" for warehouses, optimising logistics via cloud-native platforms.
BKW AG	Strong Swiss player in sustainable energy and grid modernisation.
Enel	Major European energy provider investing in renewable networks.
NextEra Energy	U.S. clean energy leader with innovative renewable grid projects.
Iberdrola	European leader in renewables and sustainable distribution networks.
Ørsted	Global leader in offshore wind and smart grid systems.
First Solar	Solar technology innovator diversifying portfolio exposure.
Siemens Energy	Technological leader in grid modernisation and renewable integration.

Company	**Weighting (%)**
Ecolab Inc (ECL)	5.48
Badger Meter Inc (BMI)	3.36
Veralto Corporation (VLTO)	5.31
Xylem Inc (XYL)	4.84
Georg Fischer AG (GF)	3.55
Waste Management Inc (WM)	5.76
GFL Environmental Inc (GFL)	3.81
Clean Harbors Inc (CLH)	3.82
Veolia Environnement (VIE.PA)	5.57
Casella Waste Systems (CWST)	4.78
Cleanaway Waste Mgmt (CWY.AX)	3.80
Tomra Systems ASA (TOM.OL)	1.95
Symbotic Inc (SYM)	1.18
GXO Logistics Inc (GXO)	2.89
Autostore Holdings Ltd (AUTO.OL)	1.91
Zebra Technologies (ZBRA)	3.46
KION GROUP AG (KGX.DE)	2.96
Cognex Corporation (CGNX)	2.84
Manhattan Associates (MANH)	3.30
BKW AG (BKW)	5.08
Enel (ENEL.MI)	6.39
NextEra Energy (NEE)	4.10
Iberdrola (IBE.MC)	6.69
Ørsted (ORSTED.CO)	2.87
First Solar (FSLR)	1.97
Siemens Energy (ENR.DE)	2.33

References

Omran, M. F., & McKenzie, E. (2000). Heteroscedasticity in stock returns data revisited: Volume versus GARCH effects [Publisher: Routledge _eprint: https://doi.org/10.1080/096031000416433]. *Applied Financial Economics*, *10*(5), 553–560. https://doi.org/10.1080/096031000416433